

Exonuclease performs the function of _____.
[2020]

- (A) Carrying the nucleus out of the cell
- (B) Disintegrating nucleus
- (C) Join the nucleotides outside nucleus
- (D) Remove nucleotides from ends of DNA

The gene transfer between unrelated plants is brought about by _____. **[2020]**

- (A) Sexual reproduction
- (B) Asexual reproduction
- (C) Cloning
- (D) Genetic engineering

The variation in temperatures in the three steps of PCR cycle is _____, _____ and _____ respectively. **[2020]**

- (A) 91°C , 55°C and 72°C
- (B) 97°C , 60°C and 92°C
- (C) 100°C , 75°C and 88°C
- (D) 95°C , 62°C and 82°C

Which one of the following is NOT a vector in genetic engineering? **[2020]**

(A) Mosquito

(B) Cosmid

(C) Plasmid

(D) Lambda phage

To introduce a desired gene into a genome
_____ is used. **[2020]**

- (A) Translation
- (B) Replication of DNA
- (C) Transcription
- (D) rDNA technology

In rDNA technology transgenic plants are prepared by using Ti plasmid of _____.

[2021]

- (A) *Agrobacterium tumefaciens*
- (B) *Bacillus thuringiensis*
- (C) *Rhizobium phaseoli*
- (D) *E. coli*

How many copies of DNA will be produced in the thermal cyclers of PCR after 5 cycles? **[2021]**

- (A) 64 (B) 16 (C) 128 (D) 32

When *Eco*RI attack palindrome and breaks each strand of DNA at site A-G phosphodiester bond it produces _____ [2022]

- (A) Staggered end
- (B) Transcribed end
- (C) Blunt end
- (D) Replicating end

The sequences recognized by restriction enzymes are _____ nucleotide long. **[2023]**

(A) 4 to 8

(B) 10 to 12

(C) 15 to 20

(D) 21 to 25

A chimeric DNA is _____.

[2019]

- (A) rDNA
- (B) replicating DNA
- (C) phage DNA
- (D) complementary DNA

In rDNA technology, the process of separation and purification of the desired final products using suitable bioreactor is collectively called _____.

[2022]

- (A) Transformation
- (B) Bio-prospecting
- (C) Upstream processing
- (D) Downstream processing

Which among the following is first transgenic plant? **[2021]**

(A) Tomato

(B) Tobacco

(C) Brinjal

(D) Cotton

Position of an organism in the environmental gradient is _____ niche. **[2023, 2021]**

(A) trophic

(B) hypervolume

(C) spatial

(D) habitat

Which of the following is **INCORRECT** statement regarding role of predators? **[2022]**

- (A) They increase the intensity of competition among competing species.
- (B) They help to maintain species diversity in a community.
- (C) Predators are prudent in nature.
- (D) They have an ability to regulate prey population.

Which among the following is NOT an example of predator and prey? **[2023]**

- (A) Frogs – locusts
- (B) Nile perch – cichlid fish
- (C) Tiger – deer
- (D) Clown fish – sea anemone

Which of the following animals are euryhaline?

I. Barnacles

II. Clams

III. *Labeo*

IV. *Catla*

Select the correct option:

[2023]

(A) I and II only

(B) II and III only

(C) I and III only

(D) III and IV only

The salinity in parts per thousand (ppt) in oceans, freshwater streams and hypersaline lagoon is _____, _____ and _____ respectively. **[2023]**

(A) 35, 100, 5

(B) 35, 5, 100

(C) 5, 35, 100

(D) 100, 5, 35

The nature and properties of soil in different places depend on **[2020]**

- (A) Climate of the place
- (B) Weathering process
- (C) Transportation of the soil
- (D) All of the above

For weakly motile organisms the immediate surrounding is called _____. **[2023, 2022]**

(A) niche

(B) microhabitat

(C) habit

(D) habitat

The sequential levels with increasing complexity of an ecological organization are
_____.

- (A) Populations → Organisms → Biomes → Communities
- (B) Biomes → Organisms → Populations → Communities
- (C) Organisms → Populations → Communities → Biomes
- (D) Organisms → Communities → Populations → Biomes